CDT204 – Computer Architecture

> Date: Nov. 6th 2025
>
> Time: 8:30 – 12:30
>
> Help: Small calculator is allowed in the exam

The exam has 54 points. The grades will be awarded as follows:

- 3 : 27 Points
- 4 : 41 Points
- 5 : 49 Points

Important Notes:

- Give as full an answer as possible to obtain full marks. All calculations, approximations, assumptions and justifications must be reported for full credit unless stated otherwise. Please use figures and examples to clarify.
- If you do not understand a question clearly, you are allowed to call the teacher and ask.
- Write the question and part number on each page clearly.
- Answer each question on a separate sheet.
- In case you might have forgotten:

  1GB = 2^{10}MB = 2^{20}KB = 2^{30}B

  1 sec = 10^3 ms = 10^6 μs = 10^9 ns = 10^{12} ps

  $$S = \frac{T_0}{\frac{T_A}{F} + (T_0 - T_A)}$$

Live long and prosper

Task 1 (9p)

It is year 8132. Global warming has taken its tolls long ago and no living creature lives on earth anymore. Some alien visitors from the planet Omicron Persei 8 have reached earth, but all they can find is an old piece of paper with a table and some text and digits on it. It contains the following:

	Total Size	Block Size	No. Ways	No. Sets	No. Bits (Tag/Index/Offset)	Cache Organization
Cache A	2^{14} Bytes	?	2	256	(? / ? / ?)	?
Cache B	32 KB	64	?	64	(? / ? / ?)	?
Cache C	? KB	64	64	1	(? / ? / ?)	?
Cache D	8 KB	?	?	?	(19/ 6/ 7)	Direct mapped

The smart person you are, you immediately realize that these are different parameters describing different cache designs. Please help the Emperor Lrrr of Omicron Persei 8 by filling the empty boxes (they have marked it with question marks for you). (Each ? 0.5p)

Note I: By "Cache Organization" we mean the term which is normally associated with the organization of information in the cache.

Task 2 (10p)

The average memory access time for a processor developed by Tyrell Corporation with 1 level of cache is 2.4 clock cycles. The properties of the memory system are as follows:

- If data is present in the cache and is valid, it can be accessed in 1 clock cycle.
- If data is not found in the cache, 80 clock cycles are needed to get it from memory.

As a designer who works for Tyrell Co. you are given the task of improving average memory access times by 60% (Meaning that the speedup in average memory access time should be 1.60). To achieve that goal you are considering adding a 2nd level of cache. This new cache has the following properties:

- The new L2 cache can be accessed in 6 clock cycles.
- The addition of this new cache does not affect the L1 cache's access patterns or hit times.
- Accessing the main memory would still require 80 additional clock cycles.

A. To obtain the desired improvement and keep your boss happy, how often must data be found in L2 cache? (in other words, what should be the hit-rate of L2 cache?) (3p)
B. What if we aim for an 80% improvement instead of the 60%? (3p)
C. A colleague of yours claims that he can improve the L1 cache in a way that it can be accessed in 1 clock cycle. How much will the average memory access time be improved by his solution assuming we do not develop the L2 at all? (3p)
D. Which solution is faster? The system that you designed for part A or the system that your colleague designed for part C? (1p)

Task 3 (6p)

Execution time can be measured as:

IC * CPI * CT

Where:

- IC, *Instruction Count*: Number of instructions in the program.
- CPI, *Cycles per Instruction:* Average number of cycles required to perform an instruction.
- CT, *Cycle Time*: Length of a clock cycle.

For each of the following design changes and for each factor in the above equation, say whether, all other things being equal, the change cannot affect the factor, will tend to increase it, or decrease it. Give a justification or explanation for your choice if you answer increase or decrease. To get full credit for any part, you need to get both the type of change, if any, and the reason correct.

a) Design change: *Adding more pipeline stages*. (Will this design change affect the total execution time? How will it affect each of IC, CPI and CT?)
b) Design change: *Larger cache*. (Will this design change affect the total execution time? How will it affect each of IC, CPI and CT?)

Task 4 (7p)

Consider the following sequence of MIPS instructions:

```
SW   R16, -100(R6)
LW   R4, 8(R16)
ADD  R5, R4, R4
```

A. Indicate dependencies and their type. (1p)
B. Assume there is no forwarding in this pipelined processor. Indicate hazards and add NOP instructions to eliminate them. (2p)
C. Assume there is full forwarding. Indicate hazards and add NOP instructions to eliminate them.(2p)

D. What is the total execution time of this instruction sequence without forwarding and with full forwarding? What is the speedup achieved by adding full forwarding to a pipeline that had no forwarding? (Assume that the latency for all the stages is equal and is 250ps for non-forwarding system and 300ps with full forwarding). (2p)

Task 5 (6p)

Suppose that you have a page size of 4 KB. Also suppose that you have an application in which the text segment (program code) takes up 2 KB, the data segment (global variables, etc.) takes up 5 KB, and that 1 KB allocated for the stack.

A. We do not know where in the process's private memory space these three segments are allocated, but we know that they are in 3 distinct address ranges. What is the maximum number of pages that this program might need? (1p)

Now assume that we know that the code segment is allocated in address 1000_{16} to $17FF_{16}$, data segment from 10000_{16} to $113FF_{16}$ and the stack from $FFC00_{16}$ to $FFFFF_{16}$ in a 1MB virtual memory. After the program runs for a while, some parts of the page table look like the following:

Valid	Dirty	Virtual Page No.	Physical Page No.
1	0	00010000	001100101
1	0	00000001	000110011
1	1	11111111	110110100
0	0	00000001	111001000
1	1	00010001	000110101

B. Considering that the virtual memory is 1MB in size, how many bits the virtual addresses should be? (1p)
C. Judging by the physical page numbers, how big is the physical memory? (1p)
D. What page number the code segment has got? (The whole segment is placed in one page) (2p)
E. Is all of the data segment read into the physical memory? (1p)

TASK 6 (8p)

A microprocessor manufacturer asks you to design a multicore processor for modern workloads. Your design contains one large core and several small cores, which share the same die. Assume the total die area is ***A*** units. The table below describes the area and performance for each core type.

Type of Core	Area (mm^2)	Performance
Large	S	$\sqrt{S}$
Small	1	1

The serial portion of a workload executes only on the large core, while the parallel portion executes on both large and small cores. On this multiprocessor, we will execute a workload where a fraction ***P*** of its work is parallel, and ***1- P*** of its work is serial. You will fit as many small cores as possible, after placing the large core. Consider the following two configurations:

- Configuration X: `A=32, S=4`
- Configuration Y: `A=32, S=16`

A. For a given workload which consists of 75% parallel tasks, which of the two configurations will be faster? How much? (4p)

B. For what values of P does the workload run faster on Y than on X? (4p)

TASK 7 (8p)

We are going to microbenchmark the cache hierarchy of a computer with the following two codes. The array `data` contains 32-bit unsigned integer values. For simplicity, we consider that access to the array `latency` bypass all caches (i.e., `latency` is not cached). `timer()` returns a timestamp in cycles. The cache hierarchy has two levels. L1 is a 4KB set associative cache.

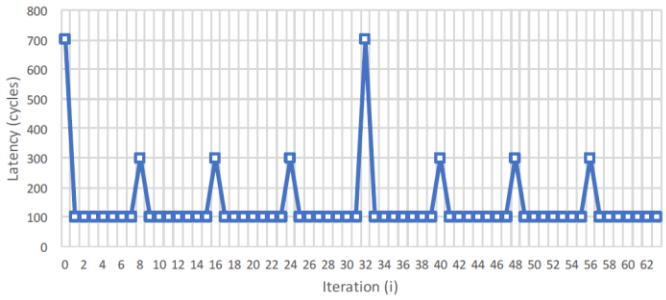
Code (1)

```
j = 0;
for (i=0; i<size; i+=stride){
    start = timer();
    d = data[i];
    stop = timer();
    latency[j++] = stop - start;
}
```

Code (2)

```
for (i=0; i<size1; i+=stride1){
    d = data[i];
}
j = 0;
for (i=0; i<size2; i+=stride2){
    start = timer();
    d = data[i];
    stop = timer();
    latency[j++] = stop - start;
}
```

A. When we run code (1), we obtain the latency values in the following chart for the first 64 reads to the array `data` (in the first 64 iterations of the loop) with `stride` equal to 1. What are the cache block sizes in L1 and L2? (4p)



B. Using code (2) with `stride1 = stride2 = 32`, `size1 = 1056`, and `size2 = 1024`, we observe `latency[0] = 300` cycles. However, if `size1 = 1024`, `latency[0] = 100` cycles. What is the maximum number of ways in L1? (4p)